

Product Data Cleaning and Analysis

Submitted By:

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Course:

Data Analytics

Project Objective

To clean, format, analyse, and visualize product data using Excel.

Dataset Description

The dataset contains product details such as manufacturing date, country code, product brand, price, quantity, and category.

Column Description

Column Name	Description
Manufacturing Date	Product manufacturing date
Country Code	Product country code
Product Brand	Product and brand name
Price (₹)	Product price
Quantity	Product quantity
Category	Product category

Data Cleaning & Transformation

The following data cleaning and transformation steps were performed:

- Checked and handled missing values in the Price column.
- Replaced missing prices with 0.
- Filled blank category values with "Unknown".
- Corrected spelling mistakes and typos in the Category column.
- Standardized text formats using PROPER, TRIM, and CLEAN functions.
- Removed duplicate records using Remove Duplicates.
- Split Product ID into Manufacturing Date and Country Code.

- Merged Brand Name and Product Name into Product Brand.
- Applied currency formatting to the Price column.
- Applied DD-MM-YYYY date formatting to Manufacturing Date.
- Applied conditional formatting to Price and Category columns.

Visualization and Insights Arrived with Screenshot of Visualization

- Applied conditional formatting to Price column
- Highlighted Electronics category in green
- Electronics products had higher prices
- Most products belonged to Electronics category

Report/Dashboard Screenshot

Manufacturing Date	Country Code	Product Brand	Price (₹)	Quantity	Category
28-01-2025	US	Dell Laptop	₹ 1,000	30	Electronics
15-02-2025	US	Nike Sneakers	₹ 80	15	Fashion
03-03-2025	US	Keurig Coffee Maker	₹ 130	40	Kitchen
11-04-2025	US	Samsung Smartphone	₹ 900	25	Electronics
22-05-2025	US	North Face Backpack	₹ 70	20	Unknown
07-06-2025	UK	Sony Headphones	₹ 0	45	Electronics
19-07-2025	UK	Adidas T-Shirt	₹ 30	5	Fashion
23-08-2025	UK	Ninja Blender	₹ 90	35	Kitchen
05-09-2025	UK	Apple Tablet	₹ 500	50	Electronics
14-10-2025	UK	Timberland Hiking Boots	₹ 130	10	Outdoor
17-06-2025	IN	HP Laptop	₹ 950	25	Electronics
25-11-2025	AU	Adidas Sneakers	₹ 90	40	Unknown
08-12-2025	DE	Nespresso Coffee Maker	₹ 120	35	Unknown
18-02-2025	CA	Fitbit Smartwatch	₹ 150	15	Electronics
16-04-2025	ES	Bose Headphones	₹ 250	20	Electronics
21-08-2025	CA	Samsonite Laptop Bag	₹ 50	35	Accessories
20-09-2025	CN	Huawei Smartwatch	₹ 160	15	Electronics
27-01-2025	IT	Asus Laptop	₹ 980	10	Electronics
01-03-2025	UK	Oakley Sunglasses	₹ 150	15	Fashion
14-08-2025	US	Coleman Camping Tent	₹ 0	10	Outdoor
14-05-2025	RU	Nikon Camera	₹ 700	50	Electronics
09-01-2025	CA	Panasonic Microwave	₹ 80	20	Kitchen
19-07-2025	BR	Xiaomi Fitness Tracker	₹ 150	30	Unknown
29-09-2025	CA	Google Smartphone	₹ 600	45	Electronics
03-06-2025	CA	Ray-Ban Sunglasses	₹ 0	25	Fashion
11-07-2025	CA	Vitamix Blender	₹ 490	40	Kitchen
07-03-2025	CA	Zara Dress	₹ 50	30	Fashion
13-04-2025	CA	Hamilton Toaster	₹ 40	10	Kitchen
24-05-2025	CA	Garmin Fitness Tracker	₹ 130	5	Electronics
02-12-2025	CA	Levi's Jeans	₹ 50	50	Fashion
09-07-2025	FR	Casio Watch	₹ 100	20	Accessories

Task	Result
Handling Missing Values – Price Column	Checked for missing values in the Price column and replaced blank values with 0.
Handling Missing Values – Category Column	Filled missing category values with "Unknown".
Correcting Inconsistent Data – Product Name	Standardized text format using PROPER, TRIM, and CLEAN functions.
Correcting Inconsistent Data – Category	Corrected spelling mistakes and typos using Find & Replace (Ctrl + H).
Find and Replace	Used Find & Replace to standardize Product Name formatting and fix category misspellings.
Removing Duplicates	Identified and removed duplicate rows using Remove Duplicates option.
Splitting Product ID	Split Product ID into Manufacturing Date and Country Code columns and removed unnecessary characters.
Merging Columns	Merged Brand Name and Product Name into a new column called Product Brand.
Number Formatting – Price	Formatted the Price column into ₹ currency format.
Number Formatting – Manufacturing Date	Formatted dates into DD-MM-YYYY format.
Conditional Formatting – Price	Applied Data Bars/Color Scales to the Price column.
Conditional Formatting – Category	Highlighted cells where Category = "Electronics" using a custom conditional formatting rule.

Descriptive Analysis

- Highest price: ₹1000
- Lowest price: ₹0
- Most common category: Electronics
- Highest quantity: 50

Diagnostic Analysis

- Missing values and typos affected data consistency
- Duplicate rows could affect analysis accuracy

Predictive Analysis

- Electronics products may continue to have higher prices
- High quantity products may have higher demand

Prescriptive Analysis

- Maintain proper category names
- Avoid missing values during data entry
- Remove duplicates regularly

Conclusion

The dataset was cleaned and analysed successfully using Excel. Missing values, duplicates, and inconsistencies were corrected. The data is now ready for further analysis and reporting.